

Short Papers

Kinematic Design for Platoon-Lane-Change Maneuvers

Harry Chia-Hung Hsu and Alan Liu

Abstract—For the lane-change maneuvers under the architectures of automated highway systems, most researchers focus on the maneuvers for a single vehicle to change lanes. However, not only are the lane-change maneuvers for a single vehicle needed, but a platoon-lane-change (PLC) maneuver is also required in some situations for an entire platoon to change lanes. The goal of this paper is to design PLC maneuvers under the coordinated- and noncoordinated-platooning infrastructures. Two PLC maneuvers are proposed in this paper, and they are the leader and predecessor PLC maneuvers, where the predecessor PLC maneuver is considered in two cases, i.e., the intraplatoon spacing less than or greater than the minimum safety spacing. We use SmartAHS to simulate these PLC maneuvers, and the simulation results demonstrate the feasibility of these PLC maneuvers while taking the safety and comfort of passengers into account.

Index Terms—Automated highway system (AHS), formation control, lane-change maneuver, platoon lane change (PLC), vehicle control.

I. INTRODUCTION

The research topic of automated highway systems (AHS) is an important research topic within the research field of intelligent transportation systems [1], [2]. The California Partners for Advanced Transit and Highways (PATH) program proposes a five-layer architecture for AHS, consisting of the physical, regulation, coordination, link, and network layers [3]–[5], and the architecture is widely used in AHS research.

Platoon control for vehicles can improve the throughput in the AHS [4]. Some typical maneuvers for longitudinal and lateral controls in a platoon are join, split, lane change, entry, exit, etc. [5], [6]. Lane-change maneuvers still remain a difficult task so far. Hatipoglu *et al.* [7] provided an analytic approach to the systematic development of controllers for lane changes. Kanaris *et al.* [8] reported strategies and their spacing requirements of lane changes under various AHS concepts. Jula *et al.* [9] presented a general algorithm to calculate the minimum longitudinal safety spacing for all vehicles involved.

Under the PATH AHS architecture, for a single vehicle in a platoon to change lanes, it needs to use split maneuvers to become a free agent and is then able to perform a lane-change maneuver [4]. Sun *et al.* [10] eliminated the restriction of a free agent and proposed a “lane change within platoons” maneuver for improving the efficiency, and this maneuver reduced the spacing for a lane change from the 60 m used in the PATH AHS architecture to 2 m.

Not only are the lane-change maneuvers for a single vehicle needed, but a platoon-lane-change (PLC) maneuver is also required in some situations for an entire platoon to change lanes. For example, when

vehicles are sorted by their destinations, all vehicles in a platoon may have the same destination [11]. In this situation, all vehicles in a platoon need to perform lane-change and exit maneuvers at the same time when the platoon approaches its destination. The maneuver for an entire platoon to change lanes is briefly mentioned as the terminology PLC maneuver in [12] and [13]. However, researchers generally focus on the lane-change maneuvers for a single vehicle, and little research focuses on the PLC maneuvers.

The main objective of this paper is to propose the PLC maneuvers in coordinated- and noncoordinated-platooning environments while taking safety and comfort into account. In the coordinated-platooning environment, the platoon leader may dictate the action sequence to be followed by the vehicles in the platoon [8]; therefore, the proposed PLC maneuver is called the leader PLC maneuver. For example, when the leader dictates to all the vehicles in the platoon to perform a lane-change maneuver with no time delay, all the vehicles will traverse the same curvature at any given time if all the vehicles have the same hardware capabilities. When an emergency is encountered during the PLC maneuver, the coordinated braking reported in [8] is used for avoiding any intraplatoon collision.

In the noncoordinated-platooning environment, in contrast, no coordination of an action sequence takes place, but the infrastructure of communication between the vehicles is still supported. In such environments, when a platoon performs a PLC maneuver, called the predecessor PLC maneuver, the front vehicle should notify the vehicle behind of the timing and force for its lane-change maneuver. Therefore, each vehicle starts to perform a lane-change maneuver with a delay of communication time. When an emergency occurs during the PLC maneuver, the braking of each vehicle is also started with a delay of communication time, and we use the concept of minimum safety spacing (MSS) to let the vehicles be free from collision [8], [9]. The MSS may be less than or greater than the intraplatoon spacing, so that we discuss these two cases as the predecessor-I PLC maneuver and the predecessor-II PLC maneuver.

II. PLC MANEUVERS

In this section, we describe the leader and the predecessor PLC maneuvers under the coordinated- and noncoordinated-platooning infrastructures.

A. Leader PLC Maneuver

Before presenting the leader PLC maneuver, we describe the platform first. In Fig. 1, it shows a highway with two lanes and four vehicles with an intraplatoon spacing of 1 m. The two bold lines are the barriers of the highway, and the thin line is the marking between the two lanes. The vehicles travel from upstream to downstream with the platoon velocity v_p in the longitudinal direction. The solid line rectangles denote physical vehicles, and the dashed rectangles are the target positions of these vehicles after executing the lane-change maneuvers. The vehicle furthest to the right is called the leader vehicle, and the other vehicles are called the follower vehicles or the followers.

The leader PLC maneuver is performed for an entire platoon to change lanes under a coordinated-platooning infrastructure. Under a coordinated-platooning infrastructure, the platoon leader will arrange the action sequence including the timing and the force to be followed

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The authors are with the Department of Electrical Engineering, National Chung Cheng University, Chia-Yi 621, Taiwan, R.O.C. (e-mail: harry_hsu@compalcomm.com; aliu@ee.ccu.edu.tw).

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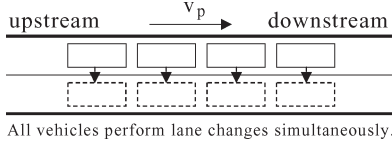


Fig. 1. Leader PLC maneuver with an intraplatoon spacing of 1 m.

by all the vehicles in the platoon [2], [8]. Based on this infrastructure, if the platoon leader dictates a lane-change maneuver with the same timing and force to all the vehicles, they will then perform the lane changes simultaneously with the same curvature at any given time, as shown in Fig. 1.

For the lane-change maneuver in this paper, we use the sinusoidal model of the lane-change maneuver from [8] to implement the lateral control. The instantaneous lateral acceleration a_y of the lane-change maneuver is as follows:

$$a_y = \frac{2\pi d_y}{t_m^2} \sin\left(\frac{2\pi t}{t_m}\right) \quad (1)$$

where d_y is the desired displacement of the lane-change maneuver (such as the lane width of 4 m), t_m is the maneuver time of this maneuver, and t is the elapsed time ($t \in [0, t_m]$). In this control, d_y is the total displacement in the lateral direction when a vehicle executes the control from the start ($t = 0$) to the end ($t = t_m$), because the double integration of a_y is the displacement of a vehicle and its value is equal to d_y ($\int_0^{t_m} \int_0^\tau a_y d\tau dt = d_y$).

Comfort is an important issue for the AHS. When the lane-change maneuver of (1) is performed, the comfort of passengers is affected by two parameters: acceleration and jerk (time derivative of acceleration) [6], [8]. During the lane-change maneuver, the longitudinal velocity is v_p at any given time, so that the longitudinal acceleration a_x and the longitudinal jerk j_x are both zeros. The peak value of the lateral acceleration is $(2\pi d_y)/t_m^2$, where it is the amplitude of a_y in (1). Moreover, the lateral jerk ($j = da/dt$) is as follows:

$$j_y = \frac{4\pi^2 d_y}{t_m^3} \cos\left(\frac{2\pi t}{t_m}\right) \quad (2)$$

where the amplitude of j_y is its peak value. We can use the maneuver time t_m to adjust the amplitudes of the acceleration and the jerk to be within the bounds of the comfortable acceleration ($a_{com} = \pm 2 \text{ m/s}^2$ in [6]) and the comfortable jerk ($j_{com} = \pm 2.5 \text{ m/s}^3$ in [6]), respectively. For example, assume that t_m is designed as 5 s and d_y is 4 m for the lane-change maneuver (see Fig. 2), the maximum acceleration of $1.01 \text{ m/s}^2 ((2 * \pi * 4)/5^2 \approx 1.01)$ is less than a_{com} . The maximum jerk peak of $1.26 \text{ m/s}^3 ((4 * \pi^2 * 4)/5^3 \approx 1.26)$ is also within the bounds of the comfortable jerk. Hence, the passengers will be under a comfortable situation while the maneuver time of the lane change is 5 s.

Besides comfort, safety is also an important issue for the AHS and is even more important than comfort. When any emergent event occurs during the execution of the PLC maneuver, all the vehicles need to be free from collision or to be guaranteed from serious injuries, which means that the impact velocity of vehicles needs to be less than the maximum allowable impact velocity [6]. Under the coordinated-platooning infrastructure, when an emergency situation occurs, coordinated braking can be performed to stop the platoon [8]. Coordinated braking means that the platoon leader will depict the braking sequence for each vehicle to follow during a braking maneuver. The leader may ask the last vehicle to brake first, then the second to the last vehicle and, finally, the leader vehicle. Similarly, the leader may also ask all the vehicles to brake simultaneously without

a time delay so that all the vehicles will keep the same intraplatoon spacing during the braking maneuver. Under this braking sequence, all the vehicles in the platoon can be guaranteed to be free from collision.

B. Predecessor PLC Maneuvers

Predecessor PLC maneuvers are designed for executing PLC maneuvers in noncoordinated-platooning environments [2], [8]. Such platooning may occur when the platoon leader cannot broadcast messages to all the vehicles or when the coordination of the action sequence is not or cannot be supported. In this environment, each vehicle performs a lane change and notifies the vehicle behind to perform a lane change, and consequently, the vehicles perform lane changes one by one with a certain delay of communication time, see Fig. 3.

Safety of the vehicles should be guaranteed during the period of the PLC maneuver, and the concept of the MSS [9] can help us in deriving the safety spacing between the vehicles. The collision between the vehicles could be classified as of angle, side-swipe, and rear-end collisions [9]. If a collision occurs in a predecessor PLC maneuver, it may be approximate to a rear-end collision but off-center.¹ Considering two neighboring vehicles in a platoon (a leader vehicle V_L and a follower vehicle V_F), the total execution time for these two vehicles to perform lane changes is $t_m + t_c$, where t_m is the maneuver time of a lane-change maneuver, as described in Section II-A. The parameter t_c is the delay of the communication time between two vehicles; in other words, the difference of the start time for the lane-change maneuvers of these two vehicles is t_c . The MSS between these two vehicles for avoiding a rear-end collision is [9]

$$S_m = \max_t \left(\int_0^t \int_0^\lambda (a_F(\tau) - a_L(\tau)) d\tau d\lambda + (v_F(0) - v_L(0))t \right) \quad (3)$$

where t is the elapsed time ($t \in [0, t_m + t_c]$), $a_L(\cdot)$ and $a_F(\cdot)$ are the respective longitudinal accelerations of V_L and V_F , and $v_L(0)$ and $v_F(0)$ are the initial longitudinal velocities of V_L and V_F , respectively.

When an emergency case occurs during the predecessor PLC maneuver, the MSS is derived as follows. Assume that these two vehicles (i.e., V_L and V_F) have the same braking capabilities [i.e., the same maximum braking deceleration (a_{min}) and braking-delay time (t_d)] and that the braking is a unit step function. Assuming that an emergency braking occurs at time t_e ($t_e \in (0, t_m + t_c)$), the leader vehicle starts to perform emergency braking at time t_e , and the maximum braking deceleration is achieved at time $t_e + t_d$. At time t_e , the leader notifies its follower of such braking, and its follower performs a braking at $t_e + t_c$ with the maximum deceleration achievement at $t_e + t_c + t_d$. We can apply these values to (3) to derive the MSS, but we use a simpler method to derive the MSS in the following. We can treat the term in the right-hand side of (3) as the displacement of the follower vehicle minus that of the leader vehicle. Assume that the leader vehicle stops at time t_s ($t_s > t_e$), and assume that the displacement of the vehicle from the time of the maximum deceleration applied, $t_e + t_d$, to the time of stop t_s is S_b so that the total displacement of the leader vehicle after t_e is $t_d * v_p + S_b$ [i.e., $\int_{t_e}^{t_s} \int_0^\lambda a_L(\tau) d\tau d\lambda + v_L(0)(t_s - t_e)$]. Similarly, the follower vehicle stops at time $t_s + t_c$, and the displacement of the follower vehicle is $(t_c + t_d) * v_p + S_b$ [i.e., $\int_{t_e}^{t_s+t_c} \int_0^\lambda a_F(\tau) d\tau d\lambda + v_F(0)(t_s + t_c - t_e)$].

¹Exactly speaking, the collision belongs to of angle collisions but approximates to a rear-end collision. In order to simplify the MSS model, we use the model of rear-end collision to analyze it.

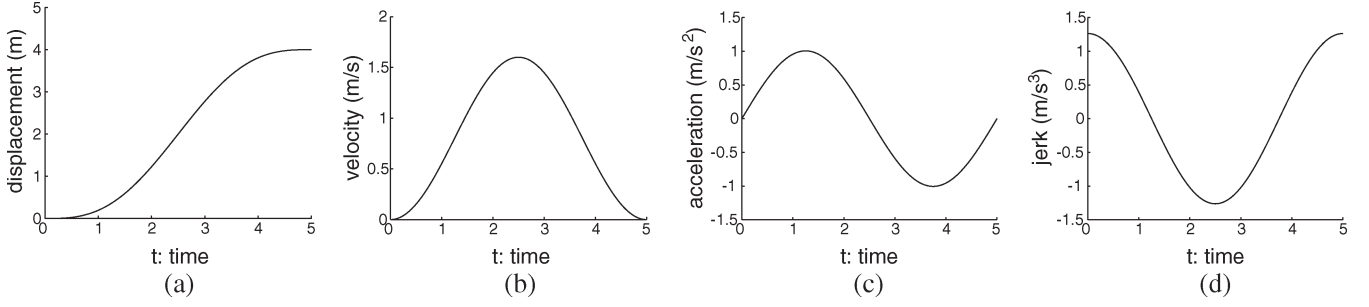


Fig. 2. Kinematic data of the lane-change maneuver. (a) Lateral displacement. (b) Lateral velocity. (c) Lateral acceleration. (d) Lateral jerk.

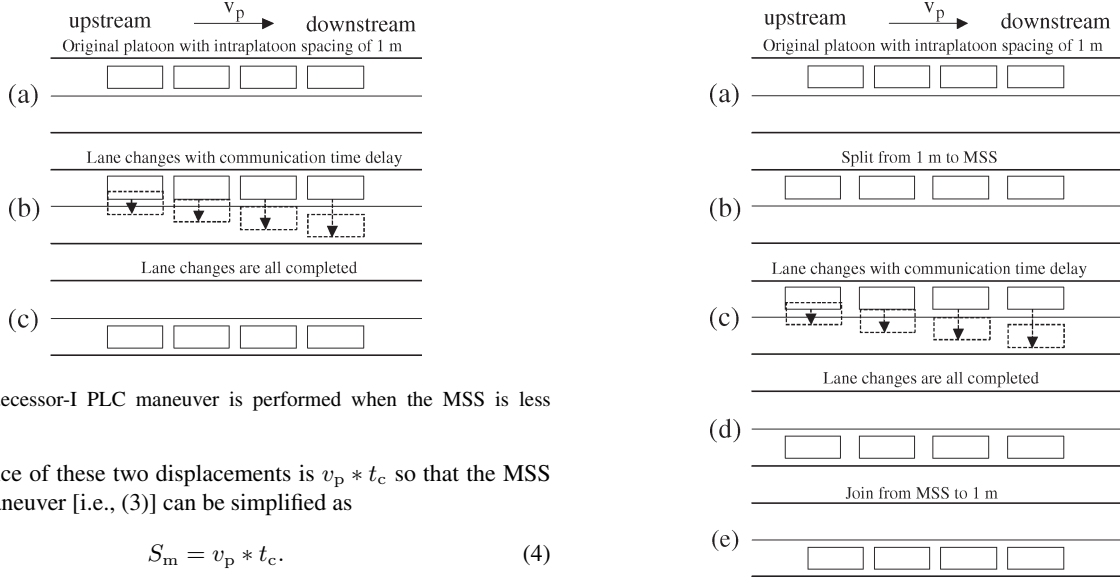


Fig. 3. Predecessor-I PLC maneuver is performed when the MSS is less than 1 m.

The difference of these two displacements is $v_p * t_c$ so that the MSS in a PLC maneuver [i.e., (3)] can be simplified as

$$S_m = v_p * t_c. \quad (4)$$

Based on (4), we can simply derive the MSS and compare whether the MSS is less than the intraplatoon spacing. If the MSS is less than the intraplatoon spacing, the predecessor PLC maneuver can be performed simply by doing lane changes one by one with a delay of communication time, as shown in Fig. 3. For example, when v_p is 20 m/s and t_c is 20 ms [6], [14], the required MSS is 0.4 m, which is less than 1 m. In contrast, if the platoon velocity is high, e.g., 30 m/s or more, and the required communication time is also large, such as 50 ms [14] or 150 ms [2], the MSS will be greater than 1 m. In this situation, the maneuver in Fig. 3 cannot guarantee the safety when an emergency occurs. In order to guarantee the safety, as shown in Fig. 4, the platoon requires to split first in order to obtain an appropriate intraplatoon spacing (greater than or equal to the MSS) and then perform lane changes one by one. After all the vehicles have changed to their target lane, all the vehicles perform join maneuvers in order to recover their original intraplatoon spacing, where the spacing is usually used in the AHS [2]. In order to distinguish these two predecessor PLC maneuvers in Figs. 3 and 4, we call them the predecessor-I PLC maneuver and the predecessor-II PLC maneuver, respectively.

Comfort and safety are important issues while performing PLC maneuvers. In the predecessor-II PLC maneuver, we can divide it into the following three stages: the split, lane change, and join stages (see Fig. 4). For the split and join stages, we adopt the split and join maneuvers from [6], [15], and [16], and the safety and comfort have been discussed and guaranteed in these publications. For the lane-change stage, the lane-change maneuver of vehicles in the predecessor-II PLC maneuver is the same as that in the leader PLC maneuver in Section II-A, so that the comfort issue has been discussed and guaranteed in Section II-A. As for the safety issue, each vehicle performs the lane-change maneuver with a spacing of at least MSS to

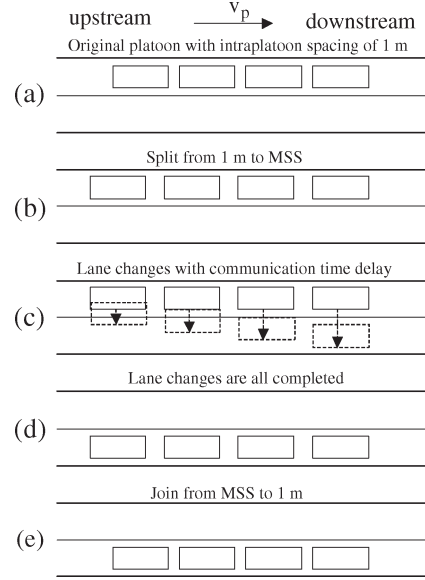


Fig. 4. Predecessor-II PLC maneuver is performed when the MSS is greater than 1 m.

its preceding vehicle, so that the safety of each vehicle is guaranteed by the work of Jula *et al.* [9]. To sum up, the comfort and safety are both guaranteed in the predecessor-II PLC maneuver, and they are also guaranteed in the predecessor-I PLC maneuver since the predecessor-I PLC maneuver is a submaneuver of the predecessor-II PLC maneuver without the split and join maneuvers.

III. SIMULATION

In this section, we use SmartAHS to simulate the normal cases and the emergency cases of the leader, predecessor-I, and predecessor-II PLC maneuvers.

A. Simulator

We used SmartAHS to build the simulator. SmartAHS [17] is a library of the SHIFT language [18] and provides a discrete and continuous (hybrid) control for the AHS. We illustrate it with the simulation results in Fig. 5. There are four snapshots in this figure, and each snapshot consists of four vehicles in the highway with two lanes. The vehicle furthest to the right is the leader vehicle, and the others are called the follower vehicles. These four vehicles are marked as V_1 – V_4 from right to left. The coordinate system in the simulation is a left-handed coordinate, in which the x -axis points toward the progress direction of the vehicles (rightwards) and the y -axis points toward the right side of the vehicles (downwards). The width of each lane in the highway is 4 m, the length of each vehicle is 5 m, and

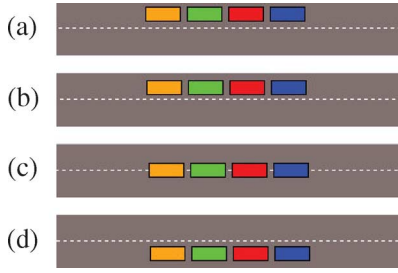


Fig. 5. Simulation results of the leader PLC maneuver, where the simulation data are listed in Table I.

TABLE I
SIMULATION DATA OF THE LEADER PLC MANEUVER

snapshot	time	x_1	y_1	v_y	a_y
A	2.00	44	2.00	0.0	0.0
B	3.25	69	2.36	0.8	1.0
C	4.50	94	4.00	1.6	0.0
D	7.00	144	6.00	0.0	0.0

the intraplatoon spacing is 1 m. The platoon travels from left to right, and the vehicle velocity in the longitudinal direction is 20 m/s. The maneuver time of the lane-change maneuver, t_m in (1), is designed as 5 s. The communication time t_c is designed as 20 ms for the predecessor-I PLC maneuver and designed as 100 ms for the predecessor-II PLC maneuver.

B. Results of Leader PLC Maneuver

Fig. 5 shows the simulation results for the four vehicles which performed the leader PLC maneuver. Table I shows the kinematic data of the simulation in Fig. 5, where time is the elapsed time, (x_1, y_1) is the position of the leader vehicle, and v_y and a_y are the lateral velocity and acceleration of these vehicles, respectively. Because the four vehicles execute the lane-change maneuvers simultaneously, their lateral velocities and accelerations are all the same, and the simulation results can also be viewed as the results executed by a single vehicle. The maneuver is started at snapshot A, with the maximum acceleration at snapshot B and the maximum lateral velocity at snapshot C, and completed at snapshot D.

C. Results of Predecessor-I PLC Maneuver

Fig. 6 shows the simulation results of the predecessor-I PLC maneuver in which MSS is 0.4 m (20 m/s * 0.02 s), which is derived by (4). The position of the leader vehicle, i.e., (x_1, y_1) , and the lateral positions of these four vehicles, i.e., y_1 to y_4 , are listed in Table II. The PLC maneuver is started at snapshot A [Fig. 6(a)]. Then, the front vehicle notifies the back vehicle one by one with the communication delay of 0.02 s; therefore, in snapshot B, all the four vehicles are executing or starting to execute lane-change maneuvers. The maximum lateral acceleration for the leader vehicle is achieved at snapshot C and the maximum velocity at snapshot D, and the maneuver is completed at snapshot E.

D. Results of Predecessor-II PLC Maneuver

We illustrate the predecessor-II PLC maneuver in Fig. 7 and Table III. The MSS is 2 m (20 m/s * 100 ms), which is derived from (4) and is greater than the intraplatoon spacing of 1 m. The PLC maneuver starts at snapshot A [Fig. 7(a)], all the four vehicles are executing the split maneuvers in snapshot B, and the split maneuvers

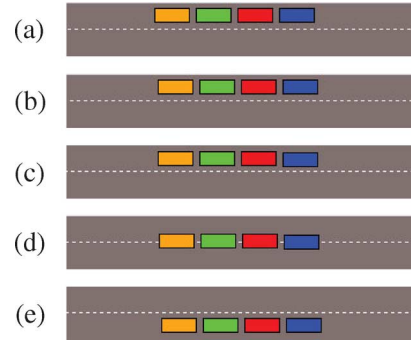


Fig. 6. Simulation results of the predecessor-I PLC maneuver, where the simulation data are listed in Table II.

TABLE II
SIMULATION DATA OF THE PREDECESSOR-I PLC MANEUVER

snapshot	time	x_1	y_1	y_2	y_3	y_4
A	2.00	44	2.00	2.00	2.00	2.00
B	2.06	45.2	2.00	2.00	2.00	2.00
C	3.25	69.0	2.36	2.29	2.22	2.17
D	4.50	94	4.00	3.97	3.94	3.90
E	7.06	145.2	6.00	6.00	6.00	6.00

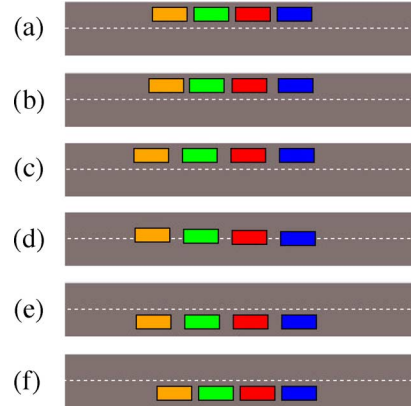


Fig. 7. Simulation results of the predecessor-II PLC maneuver, where the simulation data are listed in Table III.

TABLE III
SIMULATION DATA OF THE PREDECESSOR-II PLC MANEUVER

snapshot	time	x_1	y_1	d_{12}	d_{23}	d_{34}
A	2.00	44	2.00	1.00	1.00	1.00
B	2.30	50	2.00	1.52	1.22	0.85
C	3.26	69.2	2.00	2.00	2.00	2.00
D	5.76	119.2	4.00	2.00	2.00	2.00
E	8.57	175.4	6.00	2.00	2.00	2.00
F	9.85	201	6.00	1.00	1.00	1.00

are completed at snapshot C. At snapshot D, it is the halfway that the four vehicles perform in their lane-change maneuvers, and the lane-change maneuvers are all completed at snapshot E. This PLC maneuver is completed at snapshot F.

E. PLC Maneuvers With Emergency Braking

In order to demonstrate that the safety can be guaranteed by the leader and predecessor PLC maneuvers, we apply emergency braking to these two PLC maneuvers, respectively. For the predecessor PLC

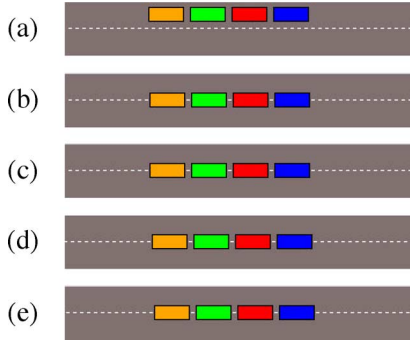


Fig. 8. Simulation results of the leader PLC maneuver with an emergency braking, where the simulation data are listed in Table IV.

TABLE IV
SIMULATION DATA OF THE LEADER PLC MANEUVER
WITH AN EMERGENCY BRAKING

snapshot	time	x_1	y_1	v_x	a_x	d
A	2.00	44	2.00	20.00	0	1.0
B	4.51	94.2	4.02	20.00	0	1.0
C	4.81	100.2	4.02	20.00	-5	1.0
D	6.80	130.1	4.02	10.05	-5	1.0
E	8.81	140.2	4.02	0.00	0	1.0

maneuver, we only apply to the predecessor-II PLC maneuver, since the predecessor-I PLC maneuver is a submaneuver and its safety can be guaranteed once the predecessor-II PLC maneuver can be guaranteed. During the braking period of a vehicle, the vehicle only applies its maximum braking (i.e., -5 m/s^2) to decelerate without changing its steering, and the comfort of the passengers is no longer considered during this period. In addition, we assume that the braking is a unit step function, and the maximum braking can be achieved after a braking-delay time of 300 ms (i.e., $t_d = 0.3 \text{ s}$).

In the simulation results of the leader PLC maneuver (Fig. 8 and Table IV), v_x and a_x are the longitudinal velocity and acceleration of these four vehicles, respectively, and d is the intervehicle distance. Between snapshots A and B in Fig. 8, the motion of the vehicles is the same as that between snapshots A and C in Fig. 5. The emergency braking is started at snapshot B, and the maximum deceleration is applied at snapshot C due to the braking-delay time of 300 ms. Snapshot D is the halfway of the braking maneuver, and all the vehicles stop at snapshot E. From the simulation results, the safety is guaranteed under the leader PLC maneuver, since d is always kept at 1.0 m.

For the predecessor-II PLC maneuver, the simulation results for applying an emergency braking during the period of the lane change are shown in Fig. 9 and Table V. Snapshots A and B in Fig. 9 are the same as snapshots A and D in Fig. 7, where vehicles performed split maneuvers first and then lane-change maneuvers. The emergency-braking command is applied one by one with a communication delay at snapshot B, and the maximum deceleration of the first vehicle is achieved at snapshot C. Snapshot D is the halfway of the deceleration, and all the vehicles stop at snapshot E. At snapshot E, d_{12} , d_{23} , and d_{34} are just zero; therefore, no collision occurs for these vehicles. To sum up, the safety of the split and join maneuvers in the predecessor-II PLC maneuver are guaranteed in [6], and the safety of the lane change is guaranteed by this simulation; therefore, the safety for the predecessor PLC maneuver is guaranteed.

IV. CONCLUSION

This paper has illustrated the PLC maneuvers for an entire platoon to change lanes while taking safety and comfort into account. Two

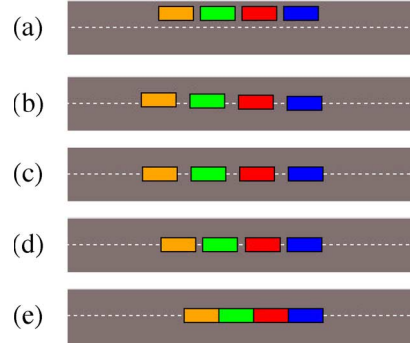


Fig. 9. Simulation results of the predecessor-II PLC maneuver with an emergency braking, where the simulation data are listed in Table V.

TABLE V
SIMULATION DATA OF THE PREDECESSOR-II PLC MANEUVER
WITH AN EMERGENCY BRAKING

snapshot	time	x_1	y_1	d_{12}	d_{23}	d_{34}
A	2.00	44	2.00	1.00	1.00	1.00
B	5.77	119.4	4.02	2.00	2.00	2.00
C	6.07	125.4	4.02	2.00	2.00	2.00
D	8.07	155.4	4.02	1.03	1.08	1.13
E	10.37	165.4	4.02	0.00	0.00	0.00

PLC maneuvers, i.e., the leader and the predecessor PLC maneuvers, were illustrated under coordinated- or noncoordinated-platooning infrastructures. When there are four vehicles in a platoon, the simulation results showed that the maneuver times are 5, 5.06, and 7.85 s for the leader, predecessor-I, and predecessor-II PLC maneuvers, respectively, in normal cases without an emergency. In addition, the simulation results also demonstrated that safety can be guaranteed when vehicles meet an emergency case. The comfort of the passengers for these three PLC maneuvers is also guaranteed by theories.

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